

Exam Date & Time: 16-Nov-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIRST SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025
ENGINEERING MECHANICS AND SMART BUILDINGS [CIV-1003]

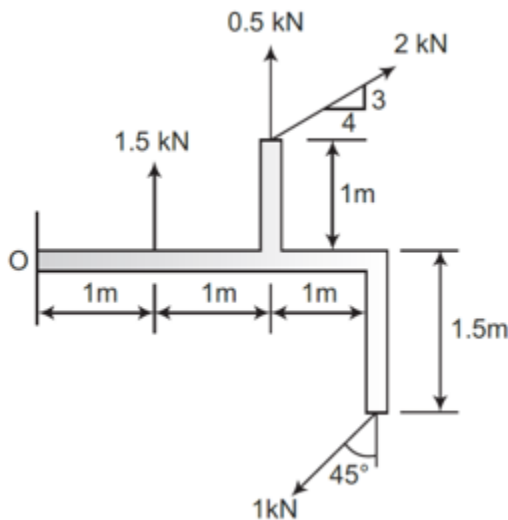
Marks: 50

Duration: 180 mins.

PART-A

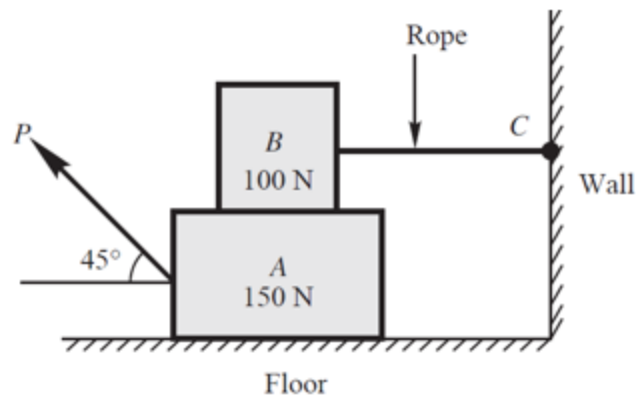
Answer all the questions.

- 1) A bracket is subjected to four forces as shown in figure.
Determine
(a) magnitude and direction of the resultant force and
A) (b) position of the resultant force from point O.



(5)

- B) Block B weighing 100 N rests on 150 N weight block A. The block B is attached to the wall by a horizontal rope BC, as shown in Figure. The coefficient of friction between A and the floor is 0.25 and between A and B is 0.25. What force P is necessary to cause motion of A to impend? (5)

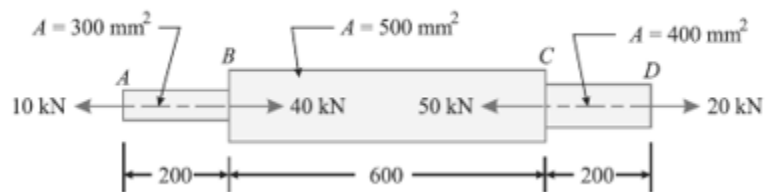


- 2) List and explain the major components of fire alarm system.

(5)

A)

- B) A member ABCD has been subjected to point loads as shown in figure. Determine the stress in each section and net change in the length of the bar. Take $E = 200 \text{ GPa}$.



(5)

- 3) Determine the value of Young's modulus and Poisson's ratio of a metallic bar of length 30 cm, breadth 4 cm and depth 4 cm when the bar is subjected to an axial compressive load of 400 kN. The decrease in length is given as 0.075 cm and increase in breadth is 0.003 cm. Also find the Bulk modulus.

(5)

A)

- B) A hole is to be punched out of a plate having an ultimate shear stress of 250 MPa. If the compressive stress in the punch is limited to 370 MPa, determine:

- (a) Maximum thickness of the plate for which a 125 mm diameter hole can be punched.
(b) If the plate is 15 mm thick, smallest diameter hole that can be punched.

(5)

- 4) Calculate the modulus of rigidity and bulk modulus of a cylindrical bar of diameter 30 mm and of length 1.5 m if the longitudinal strain in a bar during a tensile stress is four times the lateral strain. Find the change in volume, when the bar is subjected to a

(5)

- A) hydrostatic pressure of 100 N/mm^2 . Take $E = 1 \times 10^5 \text{ N/mm}^2$.

- B) A steel flat of thickness 16 mm tapers uniformly from 80 mm at one end to 50 mm at the other end in a length of 800 mm, If the flat is subjected to a load of 150 kN, find the extension of the flat. Also calculate the percentage error if average area is used for calculating its extension. Take $E = 2 \times 10^5 \text{ MPa}$.

(5)

- 5) Explain the working mechanism of Temperature sensor.

(5)

A)

B) Explain the working mechanism of Resistive Humidity Sensors (5)

-----End-----

Event Name		Subject	
058-END SEMSTER-ODD-2025		CIV-1003:ENGINEERING MECHANICS AND SMART BUILDINGS	
#	Course Outcome	Description	Marks
1	1	CO 1	5
2	2	CO 2	5
3	3	CO 3	5
4	4	CO 4	25
5	5	CO 5	10
Total			50

* NA - CO Not Selected

Event Name		Subject	
058-END SEMSTER-ODD-2025		CIV-1003:ENGINEERING MECHANICS AND SMART BUILDINGS	
#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	1	5(10%)
3	Application	9	45(90%)
4	Analysis	0	0(0%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)